

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Justification

This is the most severely misaligned dimension. The deployment requires (a) NER over regulatory entity types — INN (distinct from CHEM), ATC code, excipient name, EMA-templated posology, contraindication qualifier, safety warning — and (b) STS calibrated for regulatory equivalence sensitive to dose-threshold and contraindicated-population qualifier differences. The benchmark provides neither. NER schemas use clinical UMLS Semantic Groups (QUAERO 10 classes [Q35, Q103]; Mantra-GSC 10–11 classes [Q106]; E3C clinical entities [Q104]; DEFT2021 13 clinical types) and clinical PxCorpus 38 classes [Q44, Q108], none of which include INN-vs-CHEM distinction, ATC codes, or contraindication qualifiers (QUAERO-D16, QUAERO-D17, QUAERO-D21, MANTRAGSC-D15, PXCORPUS-D15). Dose values systematically receive O tags in QUAERO (QUAERO-D18, QUAERO-D19) and E3C (E3C-D7, E3C-D8). STS is calibrated for general semantic proximity [Q41, Q54]; CLISTER scores a 2× numeric difference (10 vs. 20 mmol/l) at 4.0 (CLISTER-D13), DEFT2020 scores a dropped 'congénitale' qualifier at 4.6 (DEFT2020-D18) and a specific-substance vs. generic-term swap at 4.3 (DEFT2020-D20). The deployment explicitly treats such differences as critical mismatches (A3). External evidence (Kadi et al. 2025 [WEB-11]) confirms ATC and excipient extraction from SmPCs is unsolved field-wide. OO is HIGH priority per the elicitation, and the fundamental decision rule for STS and the entity inventory for NER are both misaligned.

ID	Evidence
WEB-11	Kadi et al. 2025 confirms ATC codes and excipient dosages 'frequently missing or misclassified' even with state-of-the-art LLMs on EMA SmPCs

Strengths

- QUAERO CHEM class captures pharmaceutical substances including INNs and excipients at a coarse level, providing partial signal (QUAERO-D7, QUAERO-D15)
- PxCorpus NER schema includes DOSAGE, MODE, FREQUENCY, DURATION, SUBSTANCE [Q108] — the closest approach to regulatory posology entity labels
- DEFT2021 NER includes treatment, dosage, mode, and frequency entity types (DEFT2021-D5, DEFT2021-D6) — partial pharmaceutical entity coverage
- STS uses graded 0–5 scale with multiple annotators averaged [Q41], providing variance signal (DEFT2020-D10, DEFT2020-D11) even if calibration is wrong

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Benchmark NER label categories (UMLS Semantic Groups: GEOG, PHEN, DISO, ANAT, OBJC, PHYS, PROC, DEVI, CHEM, LIVB [Q103]; PxCorpus 38 clinical-prescription classes [Q108]) are calibrated for clinical research, not EU regulatory entity extraction. STS is general semantic similarity [Q54], not regulatory equivalence (CLISTER-D13).

ID	Evidence
CLISTER-D13	L'AHG urinaire est à 10 mmol/l. / L'AHG urinaire est à 20 mmol/l. — 2× numerical difference in measured value scored near-equivalent; problematic for dose-threshold regulatory use

OO-2: Missing categories specific to the regional context: INN as distinct from CHEM (QUAERO-D15, QUAERO-D17, MANTRAGSC-D15), ATC codes [WEB-11], excipient names as distinct from CHEM, EMA-templated posology expressions, contraindication qualifiers (QUAERO-D21, PXCORPUS-D15), safety warning categories. The 2025 Kadi et al. study confirms ATC and excipient extraction from SmPCs is field-wide unresolved [WEB-11].

ID	Evidence
WEB-8	EMA QRD v11 draft requires INN inclusion in SmPC Section 1 — confirms INN as central regulatory entity
WEB-11	Kadi et al. 2025 confirms ATC codes and excipient dosages 'frequently missing or misclassified' even with state-of-the-art LLMs on EMA SmPCs
QUAERO-D15	"Refludan 50 mg en poudre pour solution injectable ou pour perfusion Lepirudine Les autres composants sont le mannitol (E 421) et l' hydroxyde de sodium pour l' ajustement du pH ." — INN, brand name, and excipients all tagged as CHEM
QUAERO-D21	"Si vous avez des perturbations graves du système immunitaire (dues à une maladie , telle que leucémie ou infection à VIH" — Contraindication qualifier present but no labeled contraindication entity in schema
PXCORPUS-D15	kardegic 160 milligrammes 1 sachet par jour pendant 1 mois si absence d'ulcère — Contraindication qualifier ("si absence d'ulcère") tagged O; not in NER schema
MANTRAGSC-D15	"Effet de la testostérone sur la sécrétion de prolactine." — Both `testostérone` and `prolactine` tagged identically as CHEM; INN vs. endogenous substance distinction absent

OO-3: Categories not strictly 'non-regional' since both deployment and benchmark are in EU/French context, but UMLS Semantic Groups encode a clinical-research ontology rather than a regulatory ontology. Conflation of INN with general CHEM (QUAERO-D17) misrepresents regulatory entity distinctions critical under EMA QRD requirements [WEB-8].

ID	Evidence
QUAERO-D17	"Une cirrhose du foie peut également affecter l' excrétion rénale de la lépirudine ." — INN in pharmacokinetic sentence; CHEM label conflates API with other chemical entities
WEB-8	EMA QRD v11 draft requires INN inclusion in SmPC Section 1 — confirms INN as central regulatory entity

OO-4: Stakeholder-driven taxonomy redesign is required: regulatory-affairs experts would need to define INN/ATC/excipient/posology/contraindication-qualifier entity types and a regulatory-equivalence STS rubric calibrated for dose-threshold and population-qualifier sensitivity. No such schema exists in published French NLP literature (gap_id 3 RESOLVED as confirmed gap).

ID	Evidence
WEB-11	Kadi et al. 2025 confirms ATC codes and excipient dosages 'frequently missing or misclassified' even with state-of-the-art LLMs on EMA SmPCs

OO-5: Taxonomy issues harming structural and content validity: (a) NER schemas conflate INN, excipient, brand name under CHEM; (b) no dose/posology entity class in QUAERO despite abundant dose content; (c) no contraindication qualifier class anywhere; (d) STS scoring rubric demonstrably treats legally material differences as near-synonymy (CLISTER-D13, DEFT2020-D18, DEFT2020-D20).

ID	Evidence
QUAERO-D18	"La dose quotidienne maximale est de 21 , 6 microgrammes par jour ." — Specific dose value and unit are unannotated (O tags); dosage not a labeled entity class
E3C-D7	Isoniazide 5mg/kg, Rifampicine 10 mg/kg, Ethambutol 25 mg/kg et Pyrazinamide 30 mg/kg pendant deux mois — Confirmatory of D1: INN drugs + dosages all O-tagged
CLISTER-D13	L'AHG urinaire est à 10 mmol/l. / L'AHG urinaire est à 20 mmol/l. — 2× numerical difference in measured value scored near-equivalent; problematic for dose-threshold regulatory use

DEFT2020-D18	...contre-indiqué en cas de galactosémie congénitale..." vs. "...ne doit pas être utilisé en cas de galactosémie... — "Congénitale" qualifier dropped in target; scored 4.6 despite regulatory-material difference
DEFT2020-D20	- Ne pas utiliser chez les personnes présentant des difficultés de déglutition en raison du risque d'inhalation bronchique et de pneumopathie lipoïde." vs. "...inhalation de la paraffine liquide... — Specific substance name (paraffine liquide) vs. generic term (inhalation bronchique) scored 4.3 — regulatory-material difference treated as near-synonymy

Information Gaps

- No published STS rubric calibrated for regulatory equivalence in any language (gap_id 3 RESOLVED as confirmed field-wide gap)
- No published comparison of regulatory-affairs annotator judgments vs. biomedical NLP annotator judgments on EMA-format text

Requires Expert Verification

- Whether the deployment's regulatory-specific template layer can be configured to compensate for the absent INN/ATC/excipient entity types when running the benchmark-trained model as a foundational extraction layer
- Quantitative threshold for STS regulatory-equivalence calibration that the deployment requires

Remediation

Gap 1: NER schemas conflate INN, excipients, and brand names under CHEM; no ATC code, contraindication qualifier, or EMA-templated posology entity types; STS scoring treats legally material qualifier differences as near-synonymy.

Recommendation: Construct a deployment-side regulatory entity schema (INN distinct from CHEM, ATC, excipient, posology template fields, contraindication qualifier, safety warning) and re-annotate a held-out QUAERO EMEA sample under this schema. Develop a complementary regulatory-equivalence STS rubric with explicit rules for dose-threshold and population-qualifier mismatches, and validate the deployment's STS layer against this rubric rather than against CLISTER/DEFT2020 scores alone.

Gap 2: STS scoring rubric demonstrably treats 2× numerical differences (CLISTER-D13) and dropped legal qualifiers (DEFT2020-D18) as near-synonymy, opposite of the deployment's regulatory-equivalence requirement.

Recommendation: Do not rely on CLISTER or DEFT2020 STS scores as the primary signal for the deployment's compliance-flagging layer. Build a small adversarial regulatory-equivalence test set with paired examples differing only in dose threshold, qualifier, or population scope, and use it as a deployment-internal validation gate before relying on any STS-based compliance check.